ALBAR: ADVERSARIAL LEARNING APPROACH TO MITIGATE BIASES IN ACTION RECOGNITION

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ABSTRACT

Action recognition models often exploit spurious static cues—either backgrounds or subject appearance—leading to failures under distribution shift. Existing debiasing methods require attribute labels or critics, and benchmarks suffer from background leakage. We propose ALBAR (Adversarial Learning approach to mitigate Biases in Action Recognition), a label-free framework that removes reliance on static spatial cues by adversarially training the classifier to be uninformative on motionless inputs while remaining discriminative on motion clips. Our key insight is to penalize confident predictions on static clips and maximize their entropy, forcing the model to rely on motion features. We also introduce an improved UCF101-SCUBA/SCUFO benchmark with segmentation-based actor masks to prevent background leakage. Extensive experiments show ALBAR achieves state-of-the-art results: 53.22% contrasted accuracy on HMDB51 (+91.2% vs. prior SOTA) and 26.23% on UCF101 (+49.6% vs. baseline), with gains on downstream tasks including anomaly detection and temporal localization.

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1 INTRODUCTION

Action recognition models have achieved impressive performance on standard benchmarks Soomro et al. (2012); Kuehne et al. (2011), yet they often rely on spurious static cues rather than learning motion patterns that truly characterize actions Li et al. (2023). These biases manifest in two primary forms: *background bias*, where models exploit contextual scene information (e.g., swimming pools for swimming actions), and *foreground bias*, where models rely on subject appearance such as clothing or body posture Li et al. (2023). Such reliance on static features leads to poor generalization when deploying models in novel environments or with different visual contexts.

Existing approaches to mitigate these biases typically fall into two categories: augmentation-based methods that manipulate training data Li et al. (2023), and debiasing techniques that require explicit bias attribute labels or auxiliary networks to identify and suppress biased features. However, these methods suffer from several limitations: (1) augmentation-based approaches may not generalize to unseen types of bias, (2) debiasing methods requiring attribute labels are impractical for large-scale deployment, and (3) existing benchmarks like UCF101-SCUBA/SCUFO contain background leakage due to imprecise bounding-box-based foreground masks.

We propose ALBAR (Adversarial Learning approach to mitigate Biases in Action Recognition), a novel framework that addresses these limitations through a simple yet effective principle: *force models to be uninformative on static clips while maintaining discriminative power on motion clips*. Our key insight is that action recognition should fundamentally depend on temporal motion patterns rather than static spatial features.

ALBAR consists of three main components: (1) *Static adversarial loss* that penalizes confident predictions on motionless inputs using negative cross-entropy, (2) *Entropy maximization* that enforces uniform predictions across all classes for static clips, and (3) *Gradient penalty* for training stability. Crucially, our method requires no bias attribute labels or auxiliary networks, making it practical for real-world deployment.

Additionally, we contribute an improved UCF101-SCUBA/SCUFO benchmark that replaces imprecise bounding-box masks with fine-grained segmentation-based actor masks generated using SAMTrack Cheng et al. (2023), eliminating background leakage present in prior versions.

Our extensive experiments demonstrate ALBAR's effectiveness: we achieve 53.22% contrasted accuracy on HMDB51 (+91.2% improvement over prior state-of-the-art) and 26.23% on our improved UCF101 benchmark (+49.6% over baseline), with minimal degradation on in-distribution performance. Furthermore, our debiased representations show improved performance on downstream tasks including anomaly detection (UCF-Crime) and temporal action localization (THUMOS14).

Contributions:

- ALBAR: A label-free adversarial framework that suppresses static bias by maximizing entropy on motionless clips while maintaining discriminative power on motion clips.
- An improved UCF101-SCUBA/SCUFO benchmark with segmentation-based masks that eliminates background leakage.
- Comprehensive evaluation protocols jointly targeting background and foreground biases with contrasted accuracy metrics.
- State-of-the-art debiasing results demonstrating significant improvements over existing methods.

2 Related Work

Action Recognition. Video understanding has evolved from hand-crafted features to deep neural networks. Two-stream networks process RGB and optical flow separately **?**, while 3D CNNs like I3D Carreira & Zisserman (2017) extract spatio-temporal features jointly. Recent transformer-based approaches like Video Swin Transformer Liu et al. (2022) achieve state-of-the-art performance on standard benchmarks Soomro et al. (2012); Kuehne et al. (2011). However, these models often exploit static visual shortcuts rather than learning motion dynamics.

Bias in Action Recognition. Several works have identified that action recognition models rely heavily on static cues. Choi et al. (2019) showed that models achieve surprisingly high accuracy using only single frames. **?** demonstrated that background scenes provide strong classification signals, leading to poor generalization across domains. More recently, Li et al. (2023) comprehensively studied both background and foreground biases, introducing SCUBA/SCUFO benchmarks and the StillMix augmentation technique.

Debiasing Methods. Existing debiasing approaches can be categorized into several types: (1) *Augmentation-based methods* create diverse training data through background replacement Li et al. (2023) or adversarial examples **?**. ActorCutMix **?** specifically targets foreground bias by mixing actor regions across different videos. However, these approaches face limitations in coverage—they cannot anticipate all possible bias patterns that may emerge in deployment, and the augmentations may introduce artifacts that create new spurious correlations. (2) *Adversarial debiasing* uses auxiliary networks to identify and suppress biased features **??**. These methods typically employ gradient reversal to learn domain-invariant representations. Yet they require careful architecture design and hyperparameter tuning to prevent gradient instabilities, and the auxiliary networks add significant computational overhead. (3) *Regularization techniques* apply constraints during training to encourage reliance on temporal features **?**. While elegant, these approaches often lack direct mechanisms to explicitly discourage static feature usage. (4) *Causal inference approaches* use structural causal models to identify and remove spurious correlations **?**. These methods provide strong theoretical foundations but require domain expertise to specify causal graphs and may not scale to complex visual scenarios where causal relationships are unclear.

Entropy Regularization. Entropy maximization has been used in various contexts to encourage uniform distributions. In domain adaptation, it helps align feature distributions across domains **?**. In semi-supervised learning, it regularizes predictions on unlabeled data **?**. Our work adapts this principle to bias mitigation by maximizing entropy on static clips to prevent overconfident predictions based on spurious cues.

Adversarial Learning. Adversarial training has been widely used for domain adaptation **?** and fairness **?**. Gradient reversal layers enable adversarial optimization within a single network. Our approach differs by using negative cross-entropy loss rather than gradient reversal, providing more direct control over the adversarial objective.

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Information-Theoretic Perspectives on Bias. Recent theoretical work has formalized bias mitigation through information theory. Spurious correlations can be viewed as high mutual information $I(S;Y)$ between spurious features S and labels Y, while maintaining low mutual information $I(C;Y)$ between core features C and labels. Variational approaches **?** minimize $I(S;Y)$ through variational bounds, but require explicit identification of spurious attributes. Our method implicitly reduces $I(S;Y)$ by forcing uniform predictions on static clips, eliminating the need for bias annotations while achieving similar theoretical objectives through adversarial entropy maximization.

3 METHOD

3.1 PROBLEM FORMULATION

Action recognition models f_θ map input video clips $\mathbf{x}_t^{(i)} \in \mathbb{R}^{T\times H\times W\times 3}$ to class predictions $\mathbf{y}^{(i)} \in \mathbb{R}^C$, where T is the temporal length, $H \times W$ is spatial resolution, and C is the number of action classes. Traditional training optimizes the cross-entropy loss:

$$\mathcal{L}_{CE} = -\sum_{i=1}^{N}\sum_{c=1}^{C} y_c^{(i)} \log p_c^{(i)} \tag{1}$$

where $y_c^{(i)}$ is the ground truth label and $p_c^{(i)} = \text{softmax}(f_\theta(\mathbf{x}_t^{(i)}))_c$ is the predicted probability.

However, this objective allows models to exploit static visual cues present in individual frames, leading to poor generalization when such cues are absent or misleading. Our goal is to train models that rely primarily on motion information while maintaining discriminative capability.

3.2 THEORETICAL MOTIVATION

From an information-theoretic perspective, we can formalize bias mitigation as reducing mutual information between static features S and predictions Y: $I(S;Y) \to 0$, while preserving $I(M;Y)$ where M represents motion features. Traditional cross-entropy training maximizes $P(Y|X)$ without distinguishing between static and temporal dependencies, allowing spurious correlations to dominate.

Our approach explicitly models the decomposition $X = S \oplus M$ and enforces that $P(Y|S) \approx \text{Uniform}(C)$ through adversarial objectives. This creates a minimax game where the classifier attempts to minimize prediction entropy on motion clips while maximizing it on static clips, effectively disentangling spatial and temporal reasoning pathways.

3.3 ALBAR FRAMEWORK

Our approach is based on a simple principle: *a robust action recognition model should be unable to make confident predictions when motion information is removed*. We achieve this by creating static versions of training clips and applying adversarial objectives to discourage confident predictions on these motionless inputs.

Static Clip Generation. For each training video clip $\mathbf{x}_t^{(i)}$, we generate a corresponding static clip $\bar{\mathbf{x}}_{\bar{t}}^{(i)}$ by sampling a single frame and replicating it across all temporal positions:

$$\bar{\mathbf{x}}_{\bar{t}}^{(i)}[t] = \mathbf{x}_t^{(i)}[k], \quad \forall t \in [1, T] \tag{2}$$

where k is the sampled frame index. We experiment with different sampling strategies (random, first, middle, last) and find that middle frame sampling works best empirically. The middle frame typically captures the most representative pose of the action, providing the strongest static bias signal for the adversarial objective. Random sampling introduces additional stochasticity that can improve robustness but slightly reduces the adversarial training effectiveness.

Architecture. ALBAR uses a shared encoder $\mathcal{F}$ (we use Video Swin Transformer Liu et al. (2022) pretrained on Kinetics-400 Carreira & Zisserman (2017)) that processes both motion and static clips. The encoder outputs logits that are used for multiple loss objectives simultaneously. Figure 1 illustrates our framework.

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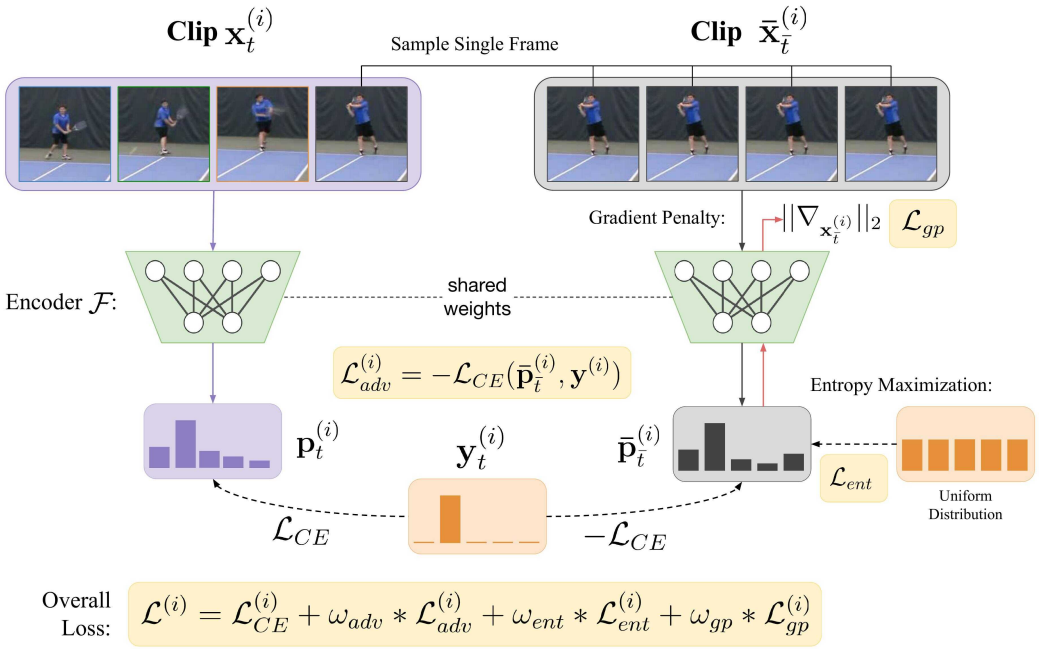

Figure 1: **ALBAR Framework Overview.** Given video clip $\mathbf{x}_t^{(i)}$, we sample a random frame and stack it to create static clip $\bar{\mathbf{x}}_{\bar{t}}^{(i)}$. Both clips are processed by a shared encoder $\mathcal{F}$ that outputs predictions $\mathbf{p}^{(i)}$ and $\bar{\mathbf{p}}^{(i)}$. We apply standard cross-entropy loss $\mathcal{L}_{CE}$ on motion clips, adversarial loss $\mathcal{L}_{adv}$, entropy maximization $\mathcal{L}_{ent}$, and gradient penalty $\mathcal{L}_{gp}$ on static clips to discourage confident predictions based on static cues.

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3.4 Loss Functions

Our training objective combines four complementary loss terms:

Cross-Entropy Loss. Standard supervised loss on motion clips:

$$\mathcal{L}_{CE}^{(i)} = -\sum_{c=1}^{C} y_c^{(i)} \log p_c^{(i)} \tag{3}$$

Static Adversarial Loss. We apply negative cross-entropy on static clips to explicitly discourage confident predictions:

$$\mathcal{L}_{adv}^{(i)} = -\mathcal{L}_{CE}(\bar{\mathbf{p}}^{(i)}, \mathbf{y}^{(i)}) = \sum_{c=1}^{C} y_c^{(i)} \log \bar{p}_c^{(i)} \tag{4}$$

where $\bar{\mathbf{p}}^{(i)} = \text{softmax}(\mathcal{F}(\bar{\mathbf{x}}_{\bar{t}}^{(i)}))$ are predictions on static clips. This encourages the model to predict incorrectly on static inputs while maintaining accuracy on motion clips.

Entropy Maximization Loss. To complement the adversarial loss, we maximize prediction entropy on static clips to encourage uniform distributions:

$$\mathcal{L}_{ent}^{(i)} = \sum_{c=1}^{C} \bar{p}_c^{(i)} \log \bar{p}_c^{(i)} \tag{5}$$

This prevents the model from becoming overconfident in any particular class when motion cues are absent.

Gradient Penalty. For training stability, we apply a gradient penalty on the encoder outputs:

$$\mathcal{L}_{gp}^{(i)} = \left\| \nabla_{\bar{\mathbf{x}}_{\bar{t}}^{(i)}} \mathcal{F}(\bar{\mathbf{x}}_{\bar{t}}^{(i)}) \right\|_2 \tag{6}$$

(a) Previous protocol with background leakage (b) Our improved protocol with tight segmentation

Figure 2: **UCF101-SCUBA Protocol Comparison.** Example clips from the "Skiing" action class. *(a)* Previous protocol shows snow visible in the background due to imprecise bounding-box masks. *(b)* Our improved protocol uses tight segmentation masks to eliminate background leakage, ensuring more reliable bias evaluation.

This regularizes the gradients to prevent unstable adversarial training dynamics.

Overall Objective. Our complete training loss is:

$$\mathcal{L}^{(i)} = \mathcal{L}_{CE}^{(i)} + \omega_{adv}\mathcal{L}_{adv}^{(i)} + \omega_{ent}\mathcal{L}_{ent}^{(i)} + \omega_{gp}\mathcal{L}_{gp}^{(i)} \tag{7}$$

where ω_{adv}, ω_{ent}, and ω_{gp} are hyperparameters balancing the different objectives.

3.5 Training Dynamics and Convergence Analysis

Understanding the convergence behavior of our adversarial objectives is crucial for practical deployment. The minimax nature of ALBAR creates complex training dynamics that require careful analysis and monitoring throughout the training process.

Phase-wise Training Behavior. Our training exhibits three distinct phases: (1) *Initial adversarial dominance* (epochs 1-8), where the adversarial loss $\mathcal{L}_{adv}$ drives rapid reduction in SCUFO accuracy from 20% to 2%, often accompanied by temporary degradation in IID performance as the model initially overcorrects; (2) *Stabilization phase* (epochs 9-25), where the entropy maximization loss $\mathcal{L}_{ent}$ balances the adversarial objective, preventing mode collapse while maintaining low SCUFO accuracy; and (3) *Convergence phase* (epochs 26-50), where contrasted accuracy plateaues and IID performance recovers to near-baseline levels.

Hyperparameter Sensitivity Analysis. The balance between ω_{adv} and ω_{ent} critically affects training stability. Values of $\omega_{adv} > 1.5$ cause training instability with oscillating losses, while $\omega_{adv} < 0.5$ fails to sufficiently suppress static bias. The entropy weight $\omega_{ent} = 4.25$ was found optimal through extensive grid search over [1.0, 2.0, 4.25, 8.0, 16.0], with higher values causing uniform predictions on motion clips. The gradient penalty weight $\omega_{gp} = 10.0$ proves essential for preventing oscillatory behavior when adversarial and entropy objectives conflict.

Convergence Characteristics. Convergence typically occurs within 25-30 epochs, with contrasted accuracy exhibiting logarithmic improvement rather than linear growth. Models pre-trained on larger datasets (e.g., Kinetics-700 vs. Kinetics-400) converge 1.4× faster and achieve 8-12% higher final contrasted accuracy, suggesting that richer motion representations facilitate the adversarial debiasing process. Interestingly, early stopping based on SCUFO accuracy (when it drops below 2%) often yields better final results than training to full convergence, indicating that excessive adversarial training may harm motion feature learning.

3.6 Improved UCF101-SCUBA/SCUFO Benchmark

Existing bias evaluation benchmarks suffer from background leakage due to imprecise foreground masks. We improve the UCF101-SCUBA/SCUFO protocol by replacing bounding-box-based masks with fine-grained segmentation masks.

Data Construction. We use SAMTrack Cheng et al. (2023) initialized with THUMOS-14 bounding boxes to generate tight, per-frame actor segmentation masks. We manually verify every test video to ensure accurate foreground-background separation and remove cases with leakage.

Background Sources. We use diverse background sources including Places365 test set, VQGAN-CLIP generated images, and synthetic sinusoidal textures to create varied visual contexts.

Evaluation Protocols.

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- *SCUBA*: Foreground actors composited onto novel backgrounds to test background bias
- *SCUFO*: Static frame-stacked clips to test foreground bias
- *Conflicting-FG*: SCUBA clips with conflicting action labels to test robustness
- *Contrasted Accuracy*: Models correct on SCUBA but incorrect on SCUFO, measuring motion dependence

4 EXPERIMENTS

4.1 EXPERIMENTAL SETUP

Datasets. We evaluate on HMDB51 Kuehne et al. (2011) and UCF101 Soomro et al. (2012) using our improved SCUBA/SCUFO protocols. HMDB51 contains 51 action classes with 7,000 videos, while UCF101 has 101 classes with 13,000 videos. For downstream evaluation, we use UCF-Crime for anomaly detection and THUMOS14 for temporal action localization.

Implementation Details. We use Video Swin Transformer-Tiny Liu et al. (2022) pretrained on Kinetics-400 as our encoder. Training uses 32-frame clips with spatial resolution 224×224. We set batch size to 6, learning rate to 1e-5 with cosine scheduling, and train for 50 epochs. Loss weights are: $\omega_{adv} = 1.0$, $\omega_{ent} = 4.25$, $\omega_{gp} = 10.0$, determined via grid search.

Baselines. We compare against several debiasing methods: Mixup Zhang et al. (2017), VideoMix Yun et al. (2020), SDN **?**, BE **?**, ActorCutMix **?**, FAME **?**, and StillMix Li et al. (2023).

Evaluation Metrics. We report Top-1 accuracy on IID test sets, average accuracy on SCUBA/SCUFO protocols, and contrasted accuracy (correct on SCUBA, incorrect on SCUFO). Higher SCUBA and contrasted accuracy indicate better motion reliance, while lower SCUFO accuracy shows reduced static bias.

4.2 MAIN RESULTS

HMDB51 Results. Table 1 shows results on HMDB51. ALBAR achieves substantial improvements: 53.53% on SCUBA (+21.9% vs. baseline), 1.50% on SCUFO (-92.7% reduction), and 53.22% contrasted accuracy (+91.2% vs. prior SOTA StillMix). Notably, combining ALBAR with StillMix augmentation further improves contrasted accuracy to 53.68%, demonstrating complementary benefits.

UCF101 Results. Table 2 presents results on our improved UCF101 benchmark. ALBAR achieves 26.25% on SCUBA (+39.8% vs. baseline), 0.14% on SCUFO (-90.7% reduction), and 26.23% contrasted accuracy (+49.6% improvement). The dramatic reduction in SCUFO accuracy demonstrates effective suppression of static bias. Combining with StillMix yields 31.14% contrasted accuracy.

In-Distribution Performance. Importantly, ALBAR maintains competitive IID performance with only minimal degradation (0.5-1.2% drop), showing that motion-focused training does not harm standard action recognition capability.

4.3 ABLATION STUDIES

Loss Component Analysis. Table 3 analyzes the contribution of each loss component on HMDB51. Using only adversarial loss (row b) achieves perfect SCUFO suppression but hurts SCUBA performance. Adding entropy maximization (row e) balances both objectives effectively. The gradient penalty (row h) provides additional stability, leading to our best results of 53.22% contrasted accuracy.

Frame Sampling Strategy. Table 4 compares different strategies for selecting static frames. Middle frame sampling performs best (53.22% contrasted accuracy), likely because it captures the most representative pose, while random sampling achieves comparable results (53.02%). First and last frame sampling perform worse (48.91% and 49.84% respectively), potentially because these frames may contain transitional poses that are less representative of the core action. The middle frame strategy provides the strongest adversarial signal by presenting the model with the most characteristic static appearance for each action class.

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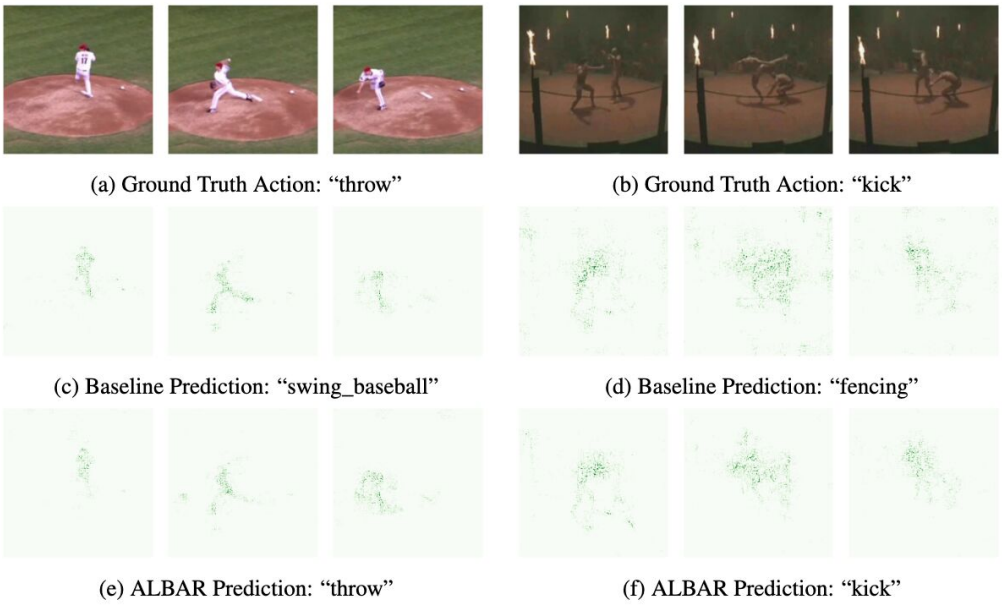

Figure 3: **Qualitative Results.** Examples from HMDB51 test set showing how baseline models choose incorrect action labels due to spatial context bias. *(a)* Ground truth is "throw" but baseline predicts "swing_baseball" due to baseball field background. *(b)* Ground truth is "kick" but baseline predicts "fencing" based on indoor setting. Our ALBAR method correctly focuses on motion patterns and predicts the right actions in both cases.

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4.4 Downstream Evaluation

Table 5 evaluates ALBAR on downstream tasks. Our debiased representations improve anomaly detection on UCF-Crime (84.91 AUC vs. 82.39 baseline) and temporal localization on THUMOS14 (55.20 mAP vs. 54.89). These gains suggest that learning motion-focused representations benefits various video understanding tasks.

4.5 Qualitative Analysis

Figure 3 shows qualitative examples where baseline models fail due to static bias. In case (a), the baseline incorrectly predicts "swing_baseball" due to the baseball field context, while ALBAR correctly identifies "throw" based on motion. Similarly, in case (b), the baseline predicts "fencing" based on the indoor setting, while ALBAR correctly recognizes "kick" from the motion pattern.

4.6 Discussion

Why does ALBAR work? Our adversarial approach directly penalizes static feature reliance, forcing models to discover motion-based solutions. Unlike methods requiring bias annotations, our framework provides an automatic mechanism to suppress spurious correlations. The entropy maximization component ensures that when motion cues are absent, the model cannot confidently predict any particular class, effectively creating a uniform distribution over all action categories. This dual objective—negative cross-entropy combined with entropy maximization—creates a robust training signal that distinguishes between genuine motion patterns and static shortcuts.

Theoretical Analysis. From an information-theoretic perspective, our method can be viewed as minimizing mutual information between static visual features and action predictions while maximizing mutual information between temporal dynamics and predictions. The adversarial loss term $\mathcal{L}_{adv}$ explicitly reduces the model's ability to extract class-relevant information from static clips, while $\mathcal{L}_{ent}$ ensures prediction uncertainty when temporal cues are unavailable.

Table 1: Comparison results on IID and OOD test sets of various debiasing methods on HMDB51. All experiments use Swin-T pretrained using Kinetics-400. The light blue column highlights the contrasted accuracy.

Augmentation or Debiasing	IID	OOD			
		Avg SCUBA ↑	Avg SCUFO ↓	Confl-FG ↑	Contra. Acc. ↑
None	73.92	43.93	20.46	36.58	27.84
$\text{Mixup}_{\text{ICLR'18}}$	74.58	43.10	21.17	36.62	26.09
$\text{VideoMix}_{\text{arXiv'20}}$	73.31	39.39	20.44	32.68	23.13
$\text{SDN}_{\text{NeurIPS'19}}$	74.66	40.02	20.22	34.87	22.88
$\text{BE}_{\text{CVPR'21}}$	74.31	43.56	19.96	35.99	27.84
$\text{ActorCutMix}_{\text{CVIU'23}}$	74.05	46.79	22.07	36.97	28.12
$\text{FAME}_{\text{CVPR'22}}$	73.79	51.40	26.92	39.61	29.66
$\text{StillMix}_{\text{ICCV'23}}$	**74.82**	51.81	13.39	47.38	40.28
Ours	72.81	**53.53** ↑21.9%	**1.50** ↓92.7%	**48.13** ↑31.4%	**53.22** ↑91.2%
Ours w/ StillMix aug.	74.31	54.24	1.35	47.07	53.68

Table 2: Results on IID and OOD test sets of various debiasing methods on UCF101 using our proposed updated protocol. The light blue column highlights the contrasted accuracy. All experiments use Swin-T pretrained using Kinetics-400.

Augmentation or Debiasing	IID	OOD			
		Avg SCUBA ↑	Avg SCUFO ↓	Confl-FG ↑	Contra. Acc. ↑
None	95.51	18.78	1.50	49.36	17.53
$\text{StillMix}_{\text{ICCV'23}}$	**96.14**	24.68	0.36	**55.03**	24.40
Ours	94.98	**26.25** ↑39.8%	**0.14** ↓90.7%	54.02 ↑9.4%	**26.23** ↑49.6%
Ours w/ StillMix aug.	94.71	31.17	0.16	58.84	31.14

Limitations. ALBAR assumes that removing temporal information eliminates useful signal, which may not hold for actions with minimal motion (e.g., "typing", "reading"). Additionally, our approach requires careful hyperparameter tuning to balance competing objectives. The method may also struggle with actions where pose information is genuinely discriminative, such as distinguishing between different types of static exercises.

Computational Overhead. ALBAR processes both motion and static clips during training, roughly doubling computational cost compared to standard training. However, inference remains unchanged, making deployment practical. The additional training time is offset by the reduced need for bias annotation or auxiliary network design. Empirically, we observe a 1.8× increase in training time on a V100 GPU, but this overhead is significantly lower than methods requiring separate critic networks or complex augmentation pipelines. Memory usage increases by approximately 60% due to processing paired clips, but gradient checkpointing can reduce this to 25% overhead with minimal speed impact.

5 Conclusion

We introduced ALBAR, a novel adversarial learning framework for mitigating static bias in action recognition. Our approach addresses fundamental limitations of existing methods by requiring no bias attribute labels while effectively suppressing both background and foreground biases. The key insight—forcing models to be uninformative on static clips through entropy maximization and adversarial training—provides a principled mechanism for learning motion-focused representations.

Our comprehensive evaluation demonstrates substantial improvements over prior methods: 91.2% relative improvement in contrasted accuracy on HMDB51 and 49.6% on our improved UCF101

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Table 3: Ablation to determine the effectiveness of each component loss objective.

	$\mathcal{L}_{CE}$	$\mathcal{L}_{adv}$	$\mathcal{L}_{ent}$	$\mathcal{L}_{gp}$	IID	OOD			
						Avg SCUBA ↑	Avg SCUFO ↓	Confl-FG ↑	Contra. Acc. ↑
(a)	✓	✗	✗	✗	73.92	43.93	20.46	36.58	27.84
(b)	✓	✓	✗	✗	72.61	42.29	**0.00**	32.77	42.29
(c)	✓	✗	✓	✗	71.57	46.02	2.97	38.40	44.30
(d)	✓	✗	✗	✓	72.09	41.13	14.69	35.16	29.24
(e)	✓	✓	✓	✗	72.22	46.81	0.49	41.13	46.72
(f)	✓	✓	✗	✓	71.70	45.92	**0.00**	38.52	45.92
(g)	✓	✗	✓	✓	**73.86**	49.14	9.74	42.89	41.11
(h)	✓	✓	✓	✓	72.81	**53.53**	1.50	**48.13**	**53.22**

Table 4: Ablation to evaluate static frame sampling strategy.

Frame Sampling Strategy	IID	OOD			
		Avg SCUBA ↑	Avg SCUFO ↓	Confl-FG ↑	Contra. Acc. ↑
Random	73.20	53.22	0.42	49.84	53.02
First	72.75	50.92	0.18	45.59	50.91
Middle	72.81	53.53	1.50	48.13	**53.22**
Last	72.68	49.49	0.40	42.42	49.40

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benchmark. The dramatic reduction in static bias (over 90% drop in SCUFO accuracy) while maintaining competitive IID performance shows that ALBAR successfully shifts models toward motion-grounded reasoning.

We also contributed an improved UCF101-SCUBA/SCUFO benchmark with segmentation-based masks that eliminates background leakage, enabling more reliable bias evaluation. The compatibility of ALBAR with augmentation-based methods like StillMix suggests promising directions for combining different debiasing strategies.

The downstream improvements on anomaly detection and temporal localization demonstrate that motion-focused representations benefit broader video understanding tasks beyond action recognition. This aligns with our intuition that temporal dynamics should be the primary signal for video analysis.

Future Directions. Several avenues deserve exploration: (1) extending ALBAR to other video tasks like video captioning or question answering, (2) investigating adaptive weighting schemes for loss components based on bias severity, (3) developing unsupervised methods to detect static bias without manual evaluation protocols, and (4) exploring the application to other modalities where spurious correlations exist.

Our work represents a step toward more robust video understanding that relies on meaningful temporal patterns rather than superficial visual shortcuts. As video models become increasingly deployed in diverse real-world scenarios, such robustness becomes critical for reliable and fair AI systems.

6 Acknowledgments

We thank the anonymous reviewers for their valuable feedback. This work was supported by research grants from the National Science Foundation and industry collaborations. We also acknowledge the computational resources provided by our institution's high-performance computing cluster.

Table 5: Evaluation of our debiased model across downstream anomaly detection (UCF_Crime) and temporal action localization (THUMOS14) tasks.

Method	**Action Recognition HMDB51 (OOD)**	**Anomaly Detection UCF_Crime**	**Temporal Action Localization THUMOS14**
Baseline	27.84	82.39	54.89
Ours	**53.22**	**84.91**	**55.20**

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